AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

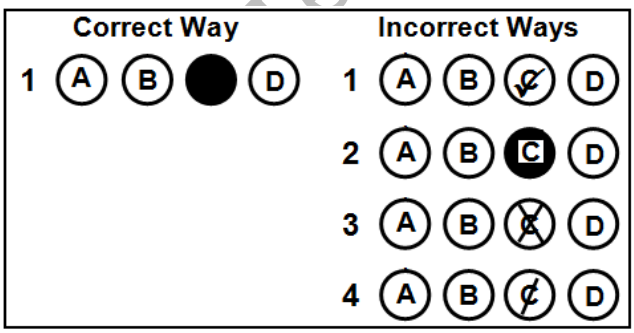
Physics Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

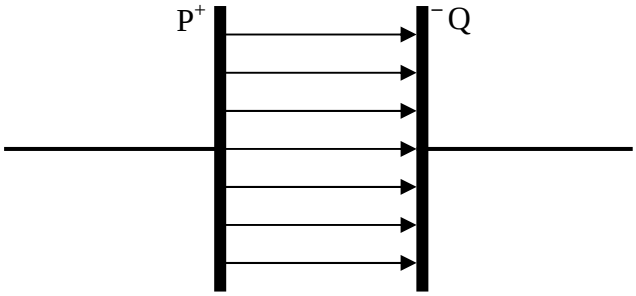
1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. In the given figure, electric lines of forces between two parallel metallic plates are directed from a positive plate (P) to a negative plate (Q).



The intensity of the electric field is

A. stronger near the plate (P).
B. stronger near the plate (Q).
C. same everywhere between the plates.
D. zero in the middle of the region between the plates.

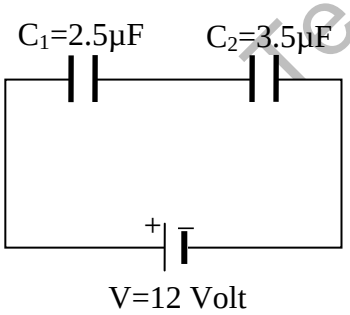
2. If a solid metallic sphere contains positive charges (Q), then according to Gauss's law, the position of charges (Q) is uniformly

A. concentrated at the centre of the sphere.
B. distributed on the inward surface of the sphere.
C. distributed on the outward surface of the sphere.
D. concentrated on the inward and outward of the sphere.

3. The total number of electric lines of force passing through a unit area is called

A. electric flux.
B. electric field.
C. electric intensity.
D. electric potential.

4. In the given circuit diagram, two capacitors are connected in series with a battery.



The equivalent capacitance of the circuit is

A. 0.69 μF
B. 1.46 μF
C. 6.00 μF
D. 8.75 μF

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5. Which of the following factors does NOT affect the resistance of a metallic conductor at constant temperature?

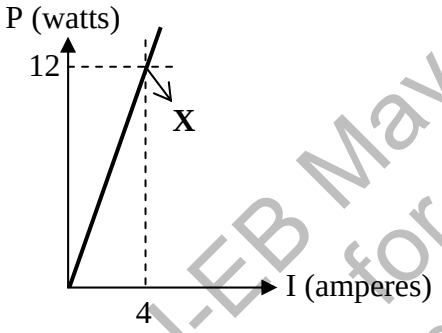
   A. Type of the conductor.
   B. Length of the conductor.
   C. Surface area of the conductor.
   D. Cross-sectional area of the conductor.

6. A 100 W bulb and a 20 W LED (light emitting diode) are connected with a 220 V power supply.

   When the power supply is turned on, then greater resistance is experienced by the

   A. 20 W LED, because it glows less bright.
   B. 20 W LED, because it draws less current.
   C. 100 W bulb, because it glows more bright.
   D. 100 W bulb, because it draws more current.

7. The given graph shows the power (P) dissipated in a variable resistor as a function of the current (I) supplied by a battery with negligible internal resistance.



   The voltage of the battery at point **X** will be

   A. 3 V
   B. 8 V
   C. 16 V
   D. 48 V

8. If 9 A current is passing through a wire of length 1.5 m and the wire experiences a perpendicular magnetic force of 0.25 N, then the magnetic field is

   A. 0.041 T
   B. 0.018 T
   C. 0.194 T
   D. 0.541 T

9. If a charged particle moving at a constant speed enters a uniform magnetic field whose direction is perpendicular to the velocity of this particle, then the particle will

   A. increase its speed.
   B. experience no force.
   C. follow a circular path.
   D. follow a parabolic path.

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10. The sensitivity of a moving coil galvanometer depends upon all the following factors EXCEPT

A. material of the pointer.
B. number of turns in the coil.
C. spring used for deflection of the needle.
D. magnetic field in which the coil is placed.

11. Which of the following phenomena occurs when an electromotive force (e.m.f.) is induced in the coil of an alternating current (AC) generator?

A. Self-induction
B. Mutual-induction
C. Electrostatic induction
D. Electromagnetic induction

12. If the magnitude of the back electromotive force (e.m.f.) in an electric motor increases with the speed of the motor, then the amount of current passing through the coil will

A. become zero.
B. remain constant.
C. gradually increase.
D. gradually decrease.

13. A transformer steps down from 220 V to 110 V. If the number of turns in the secondary coil is 40, then the number of turns in the primary coil is

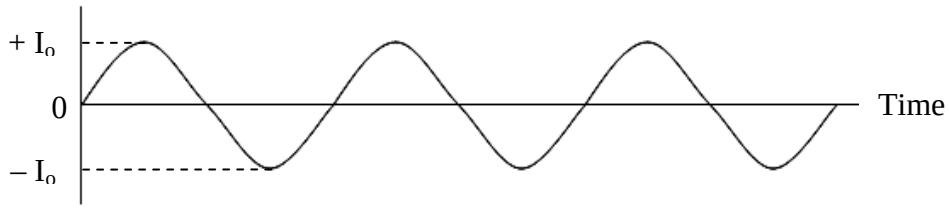
A. 20
B. 80
C. 120
D. 160

14. Which of the following electrical devices is NOT used for controlling current and voltage in an alternating current (AC) circuit?

A. Resistor
B. Inductor
C. Capacitor
D. Transistor

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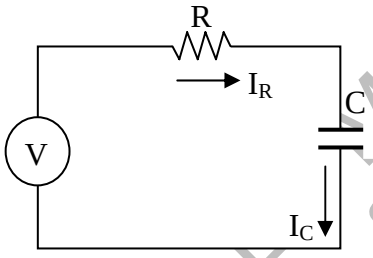
15. The variation in alternating current with time is shown by a sinusoidal wave in the given figure.



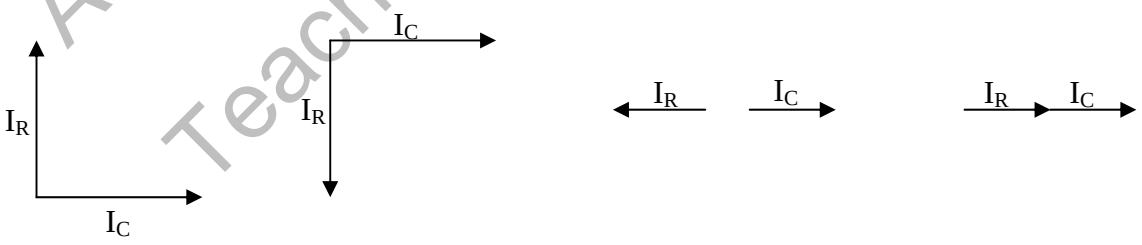
The root mean square (r.m.s.) current for this alternating current (AC) is

A. I_o
B. $\sqrt{2}\, I_o$
C. $\frac{I_o}{\sqrt{2}}$
D. $2\, I_o$

16. The given circuit diagram consists of a source of alternating voltage (V), a capacitor (C) and a resistor (R).



Which of the following vector diagrams CORRECTLY describes the phase relationship between (I_c) and (I_R)?



A B C D

17. The solid material with the highest relative elastic deformation is

A. iron.
B. glass.
C. rubber.
D. ceramic.

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18. Which of the following statements does NOT relate to an insulator?

A. Complete valence band
B. Partially filled conduction band
C. Large energy gap between electrons
D. Valence electrons are bound very tightly

19. Which of the following materials is a tetravalent semi-conductor?

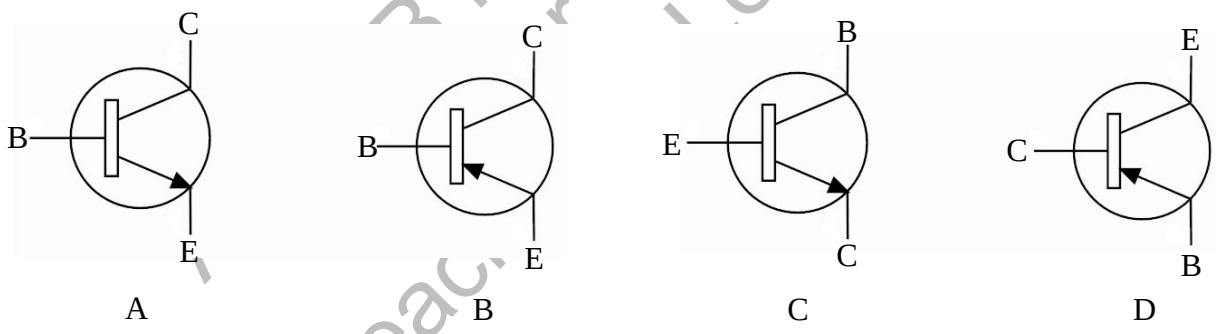
A. Barium
B. Sodium
C. Aluminium
D. Germanium

20. If two inputs (X) and (Y) of a NAND gate have 0 output, then

	X	**Y**
A	0	0
B	0	1
C	1	0
D	1	1

21. The diagram depicting p-n-p transistor is

(**Note**: Where B, C and E represent Base, Collector and Emitter respectively)



22. The scattering of X-rays by loosely bounded electrons is known as

A. compton effect.
B. pair production.
C. pair annihilation.
D. photoelectric effect.

23. Intensity of radiation emitted by a black body is directly proportional to the fourth power of temperature in Kelvin. This statement is known as the

A. Compton's effect.
B. Plank's radiation law.
C. Stefan-Boltzmann law.
D. De-Broglie's hypothesis.

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24. De-Broglie's hypothesis of wave nature of particles is applicable to

A. small masses with very low speed.
B. large masses moving with low speed.
C. large masses moving with high speed.
D. small masses moving with very high speed.

25. The series that lies in the visible region of the electromagnetic spectrum is known as

A. Balmer series.
B. Lyman series.
C. Paschen series.
D. Brackett series.

26. When an electron is removed from an atom, then the atom will become

A. excited.
B. ionised.
C. quantised.
D. partially polarised.

27. All of the following are properties of X-rays EXCEPT that they

A. are diffracted by a crystal lattice.
B. pass through wood and flesh easily.
C. are deflected by electric and magnetic fields.
D. travel in a straight line with the velocity of light.

28. Which of the following basic particles of an atom CORRECTLY corresponds the mass and charge of a positron?

	Mass of Positron	**Charge of Positron**
A	Proton	Electron
B	Electron	Proton
C	Neutron	Proton
D	Neutron	Electron

29. In an atomic reactor, the core is used to

A. protect uranium from neutrons.
B. produce heat by fission reaction.
C. absorb the fast moving neutrons.
D. accelerate the slow moving neutrons.

30. Which of the following basic forces of nature is responsible for the fusion reaction occurring in the Sun?

A. Magnetic force
B. Gravitational force
C. Weak nuclear force
D. Strong nuclear force

END OF PAPER

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